APEX RETAIL GROUP | INDEPENDENT HOMEWORK ASSIGNMENT

DOCUMENT: CLASS HOMEWORK (SESSION 3)

SCENARIO: WAREHOUSE INVENTORY THROUGHPUT & STOCKOUT RISK ANALYSIS

ASSIGNMENT OVERVIEW: You are assisting the Inventory Planning Manager at Apex Retail Group. The company operates 4 regional warehouses (North, South, East, West).

The manager needs to audit monthly stockout incidents across 6 consecutive months (January through June) to evaluate stockout frequency, variance across facilities, and inventory instability.

Standard Maximum Target: No warehouse should experience more than 5 stockouts per month.

RAW INVENTORY DATA MATRIX (4 Warehouses x 6 Months):

```
import numpy as np

# Rows: Warehouses (0: North, 1: South, 2: East, 3: West)
# Columns: Months (Jan, Feb, Mar, Apr, May, Jun)
stockout_matrix = np.array([
    [2, 4, 3, 5, 1, 3],   # North Warehouse
    [8, 12, 10, 14, 9, 11], # South Warehouse
    [1, 0, 2, 1, 3, 0],   # East Warehouse
    [6, 7, 5, 8, 9, 6]    # West Warehouse
])
```

PROBLEM 1: MATRIX INSPECTION & VECTORIZED AGGREGATIONS

In a Jupyter Notebook named `homework_session3_yourname.ipynb`:

1. Import `numpy` as `np` and initialize `stockout_matrix`.
2. Print the shape and dimensions of `stockout_matrix`.
3. Calculate and print total stockouts logged per warehouse across the 6 months (`axis=1`).
4. Calculate and print average stockouts logged per month across all warehouses (`axis=0`).
5. Calculate overall total enterprise stockouts across all warehouses and months.

PROBLEM 2: TARGET VARIANCE & DISPERSION METRICS

1. Define `max_target = 5`.
2. Compute a vectorized variance matrix: `stockout_variance = stockout_matrix - max_target`. (Positive numbers indicate SLA target breaches; negative numbers indicate compliant months).
3. Compute overall standard deviation (`np.std()`) of stockouts across the entire matrix to evaluate performance instability.
4. Use Boolean masking (`stockout_matrix > max_target`) to filter and extract all monthly stockout counts that breached the target.

PROBLEM 3: EXECUTIVE SUPPLY CHAIN MEMO

Write a 150-word formal memo to the Inventory Planning Manager containing:

1. Total enterprise stockout count and average monthly stockout rate.
2. Identification of the specific warehouse driving inventory failures and its total stockouts versus compliant facilities.
3. An explanation of how high standard deviation (dispersion) reveals systemic stockout risks in specific facilities rather than general inventory shortages.